

FLOWCHART OF AUTOMATED CONVEYOR BELT SYSTEM WITH SENSORS

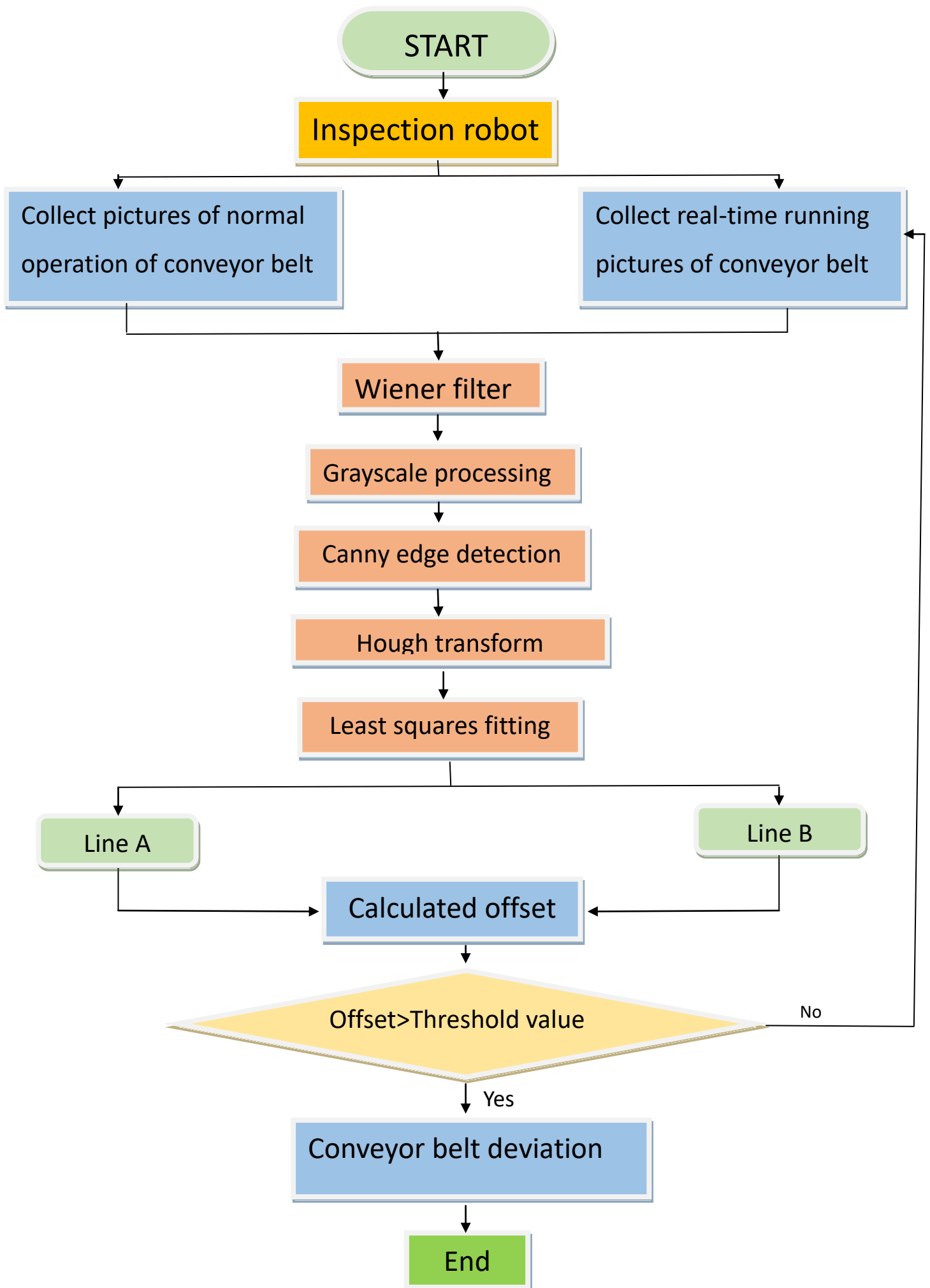


Figure 1



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WALCHAND COLLEGE OF ENGINEERING, SANGLI

(Government-Aided Autonomous Institute)

DEPARTMENT OF ROBOTICS AND AUTOMATION ENGINEERING

REPORT

Conveyor belt with sensors flowchart

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Introduction

Automation is one of the most important technologies in modern industries. It helps industries improve productivity, reduce human effort, and increase efficiency. Conveyor belt systems are widely used in manufacturing industries, warehouses, packaging units, and assembly lines for transporting materials from one location to another.

By integrating sensors and controllers, conveyor systems can automatically detect, count, and sort objects without human intervention. The Automated Conveyor Belt System with Sensors is a simple and effective example of industrial automation and Industry 4.0 applications.

This project presents the design of an automated conveyor belt system that uses sensors for object detection and sorting. The system demonstrates the principles of sensing, decision-making, and automatic material handling.

Objectives

The main objectives of this project are:

1. To design an automated conveyor belt system using sensors.
2. To detect objects automatically using an IR sensor.
3. To identify metallic and non-metallic objects.
4. To sort objects into different bins automatically.
5. To reduce manual labour and improve efficiency.
6. To understand the application of automation and Industry 4.0 concepts.

Components Required

Sr. No	Component	Function
1.	Arduino Uno/PLC	Controls the entire system
2.	IR Sensor	Detects the presence of objects
3.	Inductive Sensor	Detects metallic objects
4.	DC Motor	Drives the conveyor belt
5.	Servo Motor	Diverts objects into respective bins
6.	LCD Display	Displays object count
7.	Power Supply	Provides electrical power

Block Diagram Description

The conveyor system consists of sensors, a controller, motors, and a display unit.

IR Sensor → Controller (Arduino/PLC) → Conveyor Motor

Inductive Sensor → Controller → Servo Motor → Sorting Bin

Controller → LCD Display

The controller receives signals from sensors and performs sorting operations automatically.

Working Principle

The operation of the system is as follows:

1. The system is powered ON.
2. Sensors and motors are initialized.
3. The conveyor motor starts and moves objects continuously.
4. The IR sensor detects the presence of an object.
5. The object is counted and the count is displayed on the LCD.
6. The inductive sensor checks whether the object is metallic.
7. If the object is metallic, the servo motor diverts it to the metal bin.
8. If the object is non-metallic, it is diverted to the non-metal bin.
9. The system continuously repeats the process for incoming objects.

Flowchart

Figure 1: Flowchart of Automated Conveyor Belt System with Sensors

(Insert the professional flowchart image here.)

Applications

The Automated Conveyor Belt System with Sensors has various industrial applications:

- Manufacturing industries
- Packaging industries
- Warehouse automation
- Automobile assembly lines
- Food processing industries
- Material handling systems
- Smart factories and Industry 4.0 environments

Advantages

- Reduces human effort and labour cost.
- Improves sorting accuracy.
- Increases production speed.
- Minimizes errors in material handling.

- Provides continuous operation.
- Improves industrial productivity.
- Supports smart manufacturing systems.

Limitations

- Requires proper sensor calibration.
- Initial setup cost may be high.
- Performance depends on sensor accuracy.
- Requires periodic maintenance.

Conclusion

The Automated Conveyor Belt System with Sensors is an efficient and reliable industrial automation system for material handling and object sorting applications. The integration of sensors, controllers, and act...